Power Budget — Team 105 Motor / Rotation Control

Team Number:	105
Project Name:	**Phoenix Phorce — Handheld LiDAR Accessibility Aid**
Team Member Names	**Motor / Rotation Control — individual subsystem**
Version:	**Draft: Pololu #5138 Option 2 (motor rail TBD)**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage

All Major Components	Component Name	Part Number	yVoltageRan	#	aximumCurre	Current(m	Unit
	Scan DC gearmotor	Pololu #5138 / 1	rated (rail TE	1	670	670	mA
	Integrated encoder	Encoder on Polo	oder supply 7	1			mA
	Motor driver IC	TI DRV8833 (pac	2.7–10.8V VM	1			mA
	PIC Curiosity Nano	PIC18F57Q43	ogic rail TBD	1			mA
	Home-position sensor (if ac	TBD / switch m	TBD	1			mA
							mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power

Motor power rail — vc	Component Name	Part Number	yVoltageRan	#	aximumCurre	Current(m	Unit
	Scan DC gearmotor	Pololu #5138	6 V nominal	1	670	670	mA
	DRV8833 quiescent supply	TI DRV8833	VM TBD	1			mA
							mA
							mA
							mA
					Subtotal		mA
					Safety Margin	25%	
					Total Current Required on +12V Rail		mA
c1. Regulator or Sour	Motor supply from Power	P2 VMOTOR,	roposed / 6V	1			mA
					Total Remaining Current Available on +12V Rail		mA

Unused rail — no sep	Component Name	Part Number	yVoltageRan	#	aximumCurre	Current(m	Unit
						0	mA
						0	mA
						0	mA
						0	mA

					0		mA
				Subtotal			mA
				Safety Margin	25%		
				Total Current Required on +5V Rail			mA
c2. Regulator or Sour	Not applicable	No independen	N/A				mA
				Total Remaining Current Available on +5V Rail			mA
Unused rail — no neg	**Component Name**	**Part Number**	**yVoltageRan**	**# aximumCurre**	**Current(m**		**Unit**
							mA
					0		mA
					0		mA
					0		mA
				Subtotal			mA
				Safety Margin	25%		
				Total Current Required on -5V Rail			mA
c3. Regulator or Sour	Not applicable	No negative rai	N/A				mA
				Total Remaining Current Available on -5V Rail			mA
+3.3V local logic / enc	**Component Name**	**Part Number**	**yVoltageRan**	**# aximumCurre**	**Current(m**		**Unit**
	Curiosity Nano / PIC	PIC18F57Q43	ic (power inpu	1			mA
	Integrated encoder	Pololu #5138 en	compatibility c	1			mA
	Home sensor (if active)	TBD; omit if pass	3.3V assumed	1			mA
							mA
				Subtotal			mA
				Safety Margin	25%		
				Total Current Required on +3.3V Rail			mA
c4. Regulator or Sour	3.3V local regulator	3.3V LDO TBD —	V in / 3.3V ou	1			mA
				Total Remaining Current Available on 3.3V Rail			mA
C. For each power rail above, select a specific voltage regulator using the same process as for major							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can							
External Power Sourc	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu aximumCurre**	**Current(m**		**Unit**

Power Source 1 Selec	Shared regulated power	ower Board / P	pply current T	1		mA
Power Rails Connected to External Power	Motor rail (with margin)	Scan board + M	tor voltage T	1		mA
	3.3V LDO input load	Local logic regu	from power b	1		mA
	Other boards — total	Team-wide alloc	TBD	1		mA
	Total Remaining Current Available on External Power Source 1					mA

External Power Sourc	**Component Name**	**Part Number**	**yVoltageRan**	**Outpu aximumCurre**	**Current(m**	**Unit**
Power Source 2 Selec	Second external source i	Not selected	TBD			mA
						mA
Power Rails Connected to External Power						mA
						mA
						mA
	Total Remaining Current Available on External Power Source 2					mA

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime					
	Component Name	**Part Number**	**yVoltageRange**	**Capacity(mAh)**	**dByRegulators**
	Shared handle battery	Power Board —	y type/voltage TBD		
				Battery Life	hours

Notes

LiDAR is assigned to another PCB. Encoder is integrated in motor; count its supply current once. Driver output current is
Missing: PIC, encoder, driver, regulator current specs; 5V vs 6V motor rail; shared supply/current; battery capacity and